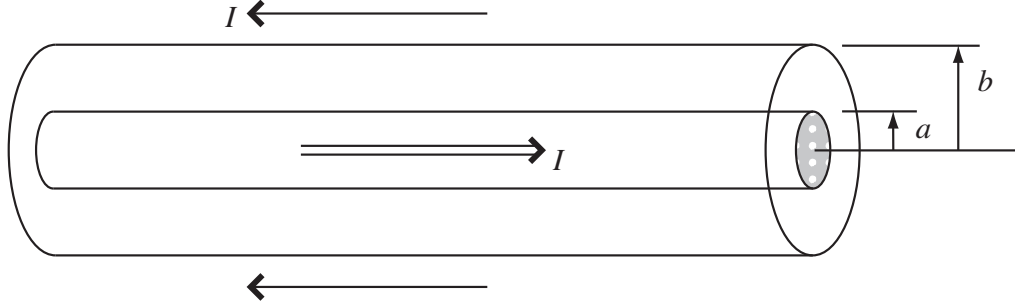


Tutorial 6: Energy, momentum, conservation laws

Question 1

A long coaxial cable carries current I down the surface of the inner cylinder (radius a) and back along the outer cylinder (radius b). Find the magnetic energy stored per unit length of cable.



[Answer: $W_B = \frac{\mu_0 I^2}{4\pi} \ln \frac{b}{a}$. Worked solution can be found in Griffith's book.]

Question 2

The two conductors of the coaxial cable of Question 1 are held at a potential difference of V . Determine the Poynting vector $\mathbf{S}$ and hence calculate the power transported down the wire.

[Answer: $P = VI$]

Question 3

- (a) Consider two equal point charges q situated on the z -axis at $z = +a$ and $-a$ (i.e. separated by a distance $2a$). In this case, since $\mathbf{p}_{\text{field}}$ is constant, the momentum conservation law can be expressed

$$\mathbf{f}_{\text{mech}} = \frac{d\mathbf{p}_{\text{mech}}}{dt} = \int_V d^3r \nabla \cdot \overleftrightarrow{T} = \oint_S \overleftrightarrow{T} \cdot d\mathbf{a}.$$

Integrate the electromagnetic stress tensor over the x - y plane to determine the force of one charge on the other.

- (b) Repeat for charges that are opposite in sign.

Question 4

A charge q moves in a magnetic field $\mathbf{B}(\mathbf{r}, t)$, making a displacement $\delta\mathbf{r}_0$ in time δt . Find the force on q from the change in the momentum of the electromagnetic field.

Hint: The $\mathbf{E}$ field comes from the charge q itself. Use the Coulomb gauge to simplify the algebra.

[Answer: $\mathbf{F} = q \frac{\delta\mathbf{r}_0}{\delta t} \times \mathbf{B}$ obviously!]

Question 5

A thin cylindrical shell of radius R and length L ($R \ll L$) carries a uniform surface charge density σ . Its angular velocity $\omega(t)\hat{z}$ smoothly increases from zero up to ω_0 . Assume that this spin up is slow enough that the displacement current $\partial\mathbf{E}/\partial t$ can be neglected.

- (a) Find the static electric field and steady magnetic field when the cylinder spins with angular velocity $\omega\hat{z}$.
- (b) Find $\mathbf{B}(\mathbf{r}, t)$ everywhere during the spin-up and use it to find the time-dependent part of the electric field.
- (c) The spin-up is achieved by an external agent (actually it was tinkerbelle) who supplies power at a rate $-\mathbf{J} \cdot \mathbf{E}$ per unit volume to the cylinder. Confirm this by evaluating Poynting's theorem over all of space. What is the role of the Poynting vector and the induced electric field in this calculation?
- (d) Evaluate Poynting's theorem over a cylindrical volume with a radius slightly smaller than the shell to study the flow of energy into the interior from the surface of the shell.

[Answer: $\pi\mu_0\sigma^2 R^4 L \omega \dot{\omega}$]

Question 6

Two concentric spherical shells carry uniformly distributed charges $+Q$ (at radius a) and $-Q$ (at radius $b > a$). They are immersed in a uniform magnetic field $\mathbf{B} = B_0\hat{z}$.

- (a) Find the angular momentum of the fields with respect to the centre.
- (b) The magnetic field is gradually turned off. Find the torque on each sphere and the resulting angular momentum of the system.

[Answer: $-\frac{Q}{3}B_0(b^2 - a^2)\hat{z}$]